

My way of using AI in theoretical chemistry research



# Accelerating Theoretical Derivations w/ Claude

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# Outline

**00 Preliminary**

**01 Reading papers**

**02 Theoretical derivations**

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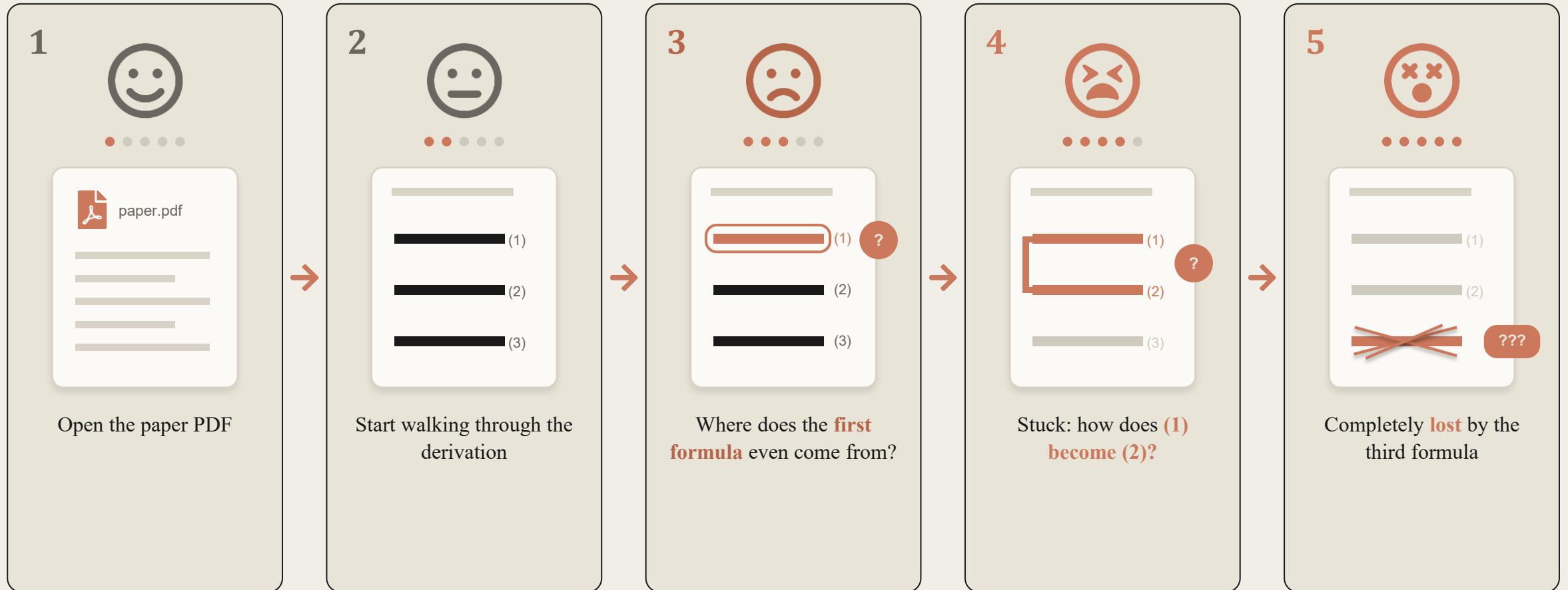
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**Preliminary**

# ■ **Daily Life as a Theoretical Chemist**

- Reading and walking through theoretical papers
- Math derivations
- Coding & numerical implementation
- Writing papers

# ■ Walking through a Paper's Derivations



**AI can bridge this gap!**

- **My Perspective on Theoretical Work**
  - Understanding is a central part of your research, AI cannot replace your own critical thinking
  - The derivation itself is your "result"—it cannot be fully captured or visualized by Web/GUI
  - Consequently, you may not be able to understand the results intuitively
  - Unlike working with standard frontend applications, manual verification is an essential and necessary component when collaborating with AI
  - This talk focuses on learning with AI, rather than merely trusting AI outputs

# 01



## Reading Papers

# Reading Old Papers

- Old-school notation conventions differ significantly from modern ones
- Optical Character Recognition (OCR) software struggles to process these types of papers
- Handling mathematical symbols is particularly difficult in this format
- There are differing conventions for mathematical operations

If the rotating system of axes has an angular velocity  $\omega$  and if the vector  $\mathbf{r}_i$  is defined as the vector with components  $x_i, y_i, z_i$  in the moving system, then the velocity of the  $i$ th particle in space is<sup>4</sup>

$$\dot{\mathbf{R}} + \omega \times \mathbf{r}_i + \mathbf{v}_i. \quad (1)$$

Consequently the kinetic energy  $T$  of the whole system is given by

$$2T = \dot{\mathbf{R}}^2 \sum m_i + \sum m_i (\omega \times \mathbf{r}_i) \cdot (\omega \times \mathbf{r}_i) + \sum m_i v_i^2 + 2\dot{\mathbf{R}} \cdot \sum m_i \mathbf{v}_i + 2\dot{\mathbf{R}} \times \omega \cdot \sum m_i \mathbf{r}_i + 2\omega \cdot \sum m_i \mathbf{r}_i \times \mathbf{v}_i. \quad (2)$$

We have not yet defined the rotating system. Following Eckart,<sup>5</sup> we do so by introducing the conditions

$$\sum m_i \mathbf{v}_i = 0 \quad \text{and} \quad \sum m_i \mathbf{a}_i \times \mathbf{v}_i = 0. \quad (3)$$

The first of these implies also that  $\sum m_i \mathbf{r}_i = 0$ , or that the origin of the moving system is at the center of gravity of the molecule. The second condition states that there is no angular momentum, to the first approximation (i.e.,  $\mathbf{a}_i$  replacing  $\mathbf{r}_i$ ), relative to the moving system.

Replacing  $\mathbf{r}_i$  by  $\mathbf{a}_i + \mathbf{q}_i$  in the last term of Eq. (2) and using the conditions (3), we obtain

$$2T = \dot{\mathbf{R}}^2 \sum m_i + \sum m_i (\omega \times \mathbf{r}_i) \cdot (\omega \times \mathbf{r}_i) + \sum m_i v_i^2 + 2\omega \cdot \sum m_i \mathbf{q}_i \times \mathbf{v}_i. \quad (4)$$

The first term is the translational energy of the system and will be omitted hereafter because it can always be separated from the other terms. The second term is the rotational energy, the third the vibrational energy, and the last the coupling between rotation and vibration. By standard methods,<sup>4</sup> the kinetic energy expression above can be reduced to the form

$$2T = A\omega_x^2 + B\omega_y^2 + C\omega_z^2 - 2D\omega_x\omega_y - 2E\omega_y\omega_z - 2F\omega_z\omega_x + \sum m_i v_i^2 + 2\omega_x \sum m_i (\mathbf{q}_i \times \mathbf{v}_i)_x + 2\omega_y \sum m_i (\mathbf{q}_i \times \mathbf{v}_i)_y + 2\omega_z \sum m_i (\mathbf{q}_i \times \mathbf{v}_i)_z, \quad (5)$$

in which  $A, B, C$  are the instantaneous moments

<sup>4</sup> For a closely similar treatment of rigid bodies which gives more of the intermediate steps see L. Page, *Introduction to Theoretical Physics*, Chapter II.

of inertia with respect to the moving  $x, y, z$  axes, respectively;  $D, E$ , and  $F$  are the products of inertia with respect to the  $x$  and  $y, y$  and  $z, z$  and  $x$  axes, respectively;  $\omega_x, \omega_y, \omega_z$  are the components of the angular velocity  $\omega$  of the rotating system of axes. It is to be emphasized that  $A, B, C$ , etc., are not constants, but are functions of the positions of the particles which change as the molecule vibrates.

## Introduction of normal coordinates

It is now convenient to introduce a set of  $3N-6$  internal coordinates, which are best chosen to be the normal coordinates  $Q_k$  of the vibration problem. These are defined in terms of the components  $\xi_i/(m_i)^{1/2}, \eta_i/(m_i)^{1/2}$ , and  $\zeta_i/(m_i)^{1/2}$  of the displacement vector  $\mathbf{q}_i$  by the equations

$$\xi_i = \sum_k l_{ik} Q_k, \quad \eta_i = \sum_k m_{ik} Q_k, \quad \zeta_i = \sum_k n_{ik} Q_k, \quad (6)$$

in which  $\xi_i, \eta_i, \zeta_i$  measure the displacement from the equilibrium position in terms of a mass-adjusted scale, and  $l_{ik}, m_{ik}, n_{ik}$  are constant coefficients.

Consequently, since the normal coordinates are normalized and orthogonal,

$$\sum m_i x_i^2 = \sum (\dot{\xi}_i^2 + \dot{\eta}_i^2 + \dot{\zeta}_i^2) = \sum_k \dot{Q}_k^2, \quad (7)$$

and

$$\begin{aligned} \sum m_i (\mathbf{q}_i \times \mathbf{v}_i)_x &= \sum (\eta_i \dot{\xi}_i - \dot{\zeta}_i \eta_i) = \sum_k \mathfrak{X}_k \dot{Q}_k, \\ \sum m_i (\mathbf{q}_i \times \mathbf{v}_i)_y &= \sum \mathfrak{Y}_k \dot{Q}_k, \\ \sum m_i (\mathbf{q}_i \times \mathbf{v}_i)_z &= \sum \mathfrak{Z}_k \dot{Q}_k, \end{aligned} \quad (8)$$

in which

$$\begin{aligned} \mathfrak{X}_k &= \sum_{i,l} (n_{il} m_{il} - m_{il} n_{li}) Q_l, \\ \mathfrak{Y}_k &= \sum_{i,l} (l_{il} n_{li} - n_{il} l_{li}) Q_l, \\ \mathfrak{Z}_k &= \sum_{i,l} (m_{il} l_{li} - l_{il} m_{li}) Q_l. \end{aligned} \quad (9)$$

The kinetic energy is therefore

$$2T = A\omega_x^2 + B\omega_y^2 + C\omega_z^2 - 2D\omega_x\omega_y - 2E\omega_y\omega_z - 2F\omega_z\omega_x + 2\omega_x \sum_k \mathfrak{X}_k \dot{Q}_k + 2\omega_y \sum_k \mathfrak{Y}_k \dot{Q}_k + 2\omega_z \sum_k \mathfrak{Z}_k \dot{Q}_k + \sum_k \dot{Q}_k^2. \quad (10)$$

- **Reading Old Papers**
  - Let Claude reframe the notation
  - Tell Claude to rewrite specific equations using modern notation
  - Prepare your own notation convention (e.g., by constructing a notation table)
  - Clarify definitions for mathematical operations (e.g., Fourier transform definitions)
  - Generate a LaTeX file as your output

# ■ Notation Table

- LaTeX, Markdown, etc
- I strongly recommend requiring AI to use **SI units** rather than atomic units or Gaussian units
- This table can be utilized in your research

| Symbol                          | Meaning                                                    |
|---------------------------------|------------------------------------------------------------|
| $i$                             | Electron indices                                           |
| $j$                             | Nuclear indices                                            |
| $N_{\text{elec}}$               | Number of electrons                                        |
| $N_{\text{nucl}}$               | Number of nuclei                                           |
| $Z_j$                           | Atomic number of nucleus $j$                               |
| $M_j$                           | Mass of nucleus $j$                                        |
| $m_e$                           | Electron mass                                              |
| $e$                             | Elementary charge                                          |
| $\hbar$                         | Reduced Planck constant                                    |
| $\epsilon_0$                    | Vacuum permittivity                                        |
| $\mathbf{r}_i$                  | Electron position                                          |
| $\mathbf{R}_j$                  | Nuclear position                                           |
| $r_{ij}$                        | Electron–electron distance $ \mathbf{r}_i - \mathbf{r}_j $ |
| $ \mathbf{r}_i - \mathbf{R}_j $ | Electron–nucleus distance                                  |
| $R_{ij}$                        | Nucleus–nucleus distance $ \mathbf{R}_i - \mathbf{R}_j $   |
| $\mathbf{x}_i$                  | Space–spin coordinate $(\mathbf{r}_i, \sigma_i)$           |

| Symbol         | Meaning / expression                                                                                                    |
|----------------|-------------------------------------------------------------------------------------------------------------------------|
| $\hat{H}$      | Full molecular Hamiltonian (non-relativistic)                                                                           |
| $\hat{T}_e$    | Electronic kinetic energy    SI: $-\sum_i \frac{\hbar^2}{2m_e} \nabla_i^2$                                              |
| $\hat{T}_n$    | Nuclear kinetic energy    SI: $-\sum_I \frac{\hbar^2}{2M_j} \nabla_{\mathbf{R}_j}^2$                                    |
| $\hat{V}_{ee}$ | Electron–electron repulsion    SI: $\frac{1}{2} \frac{1}{4\pi\epsilon_0} \sum_{i \neq j} \frac{e^2}{r_{ij}}$            |
| $\hat{V}_{en}$ | Electron–nucleus attraction    SI: $-\frac{1}{4\pi\epsilon_0} \sum_{i,j} \frac{Z_j e^2}{ \mathbf{r}_i - \mathbf{R}_j }$ |
| $\hat{V}_{nn}$ | Nucleus–nucleus repulsion    SI: $\frac{1}{2} \frac{1}{4\pi\epsilon_0} \sum_{i \neq j} \frac{Z_i Z_j e^2}{R_{ij}}$      |

# 02



## Theoretical Derivations

# ■ **Derivation Steps**

- Define your theoretical framework (e.g., in Quantum Mechanics or Quantum Field Theory)
- Define your system Hamiltonian
- Starting from your Hamiltonian, derive the formula step by step
- Clearly present the approximations you made
- Ask follow-up questions to clarify each formula

# ■ **My Prompt**

- In my prompt, there are at least three distinct parts:
  - Goals or Objectives
  - Context & Roadmap
  - Constraints & Requirements
- Divide a single, large derivation into several smaller steps
- Don't make the AI do too much in a single response; instead, have it stop at an intermediate checkpoint

### ### Objective

Provide a mathematically rigorous, comprehensive, and highly detailed step-by-step derivation of the transformation of a **classical molecular Hamiltonian from Cartesian coordinates** to **nuclear center-of-mass coordinates** within quantum mechanics.

*beginning*

*ending*

### ### Context & Roadmap

This derivation is part of a larger 4-step framework to transform a classical Cartesian Hamiltonian into a quantum Hamiltonian in center-of-mass, rotational, and internal coordinates:

1. Formulate the classical molecular Hamiltonian in Cartesian coordinates using explicit positions and conjugate momenta.
2. Transform the classical molecular Hamiltonian from Cartesian coordinates to nuclear center-of-mass and relative coordinates.
3. Transform the nuclear center-of-mass frame Hamiltonian into rotational (body-fixed) and internal (vibrational) coordinates.
4. Quantize the resulting classical Hamiltonian into its quantum mechanical operator equivalent.

### ### Scope for This Conversation

For this response, you must execute **\*\*Steps 1 and 2 only\*\***. Ensure that the derivation in Step 2 is fully explicit so that it naturally establishes the foundation for subsequent rotational and internal coordinate transformations.

### ### Strict Constraints & Requirements

1. **\*\*Mathematical Rigor:\*\*** Do not skip intermediate algebraic steps or use phrases like "it can be easily shown that." Explicitly show the coordinate transformation relations, the application of the chain rule (or velocity/momenta transformations), and how the kinetic energy terms separate into center-of-mass and relative contributions.
2. **\*\*Notation Alignment:\*\*** You must strictly follow the notation schema specified in the attached .tex file. Define every symbol clearly when it is first introduced, ensuring absolute consistency with the provided table.
3. **\*\*Formatting:\*\*** Use standard, clean LaTeX formatting for all mathematical expressions and equations.

## ■ Use Agent Skills

- Tell Claude to generate Agent Skills in your style
- Label each major result as (D) derived, (P) postulated, (A) approximated, (Def) definitional
- With these labels, you can first check *postulated* and *approximated* ones

# ■ Use Ag

■ Tell Cla

■ Label ea  
(Def) de

■ With the

## Step 1 — Classical molecular Hamiltonian in Cartesian coordinates

**1.1 Lagrangian (P).** Non-relativistic kinetic energies are quadratic in velocities; all interactions are instantaneous Coulomb. The Cartesian Lagrangian is

$$L = \underbrace{\frac{1}{2} \sum_i m_e |\dot{\mathbf{r}}_i|^2}_{\text{electronic KE}} + \underbrace{\frac{1}{2} \sum_j M_j |\dot{\mathbf{R}}_j|^2}_{\text{nuclear KE}} - V(\{\mathbf{r}\}, \{\mathbf{R}\}),$$

with the Coulomb potential (def)

$$V = \frac{1}{2} k \sum_{i \neq i'} \frac{e^2}{r_{ii'}} - k \sum_{i,j} \frac{Z_j e^2}{|\mathbf{r}_i - \mathbf{R}_j|} + \frac{1}{2} k \sum_{j \neq k} \frac{Z_j Z_k e^2}{R_{jk}}.$$

**1.2 Canonical momenta (D).** By definition  $\mathbf{p}_i = \partial L / \partial \dot{\mathbf{r}}_i$ ,  $\mathbf{P}_j = \partial L / \partial \dot{\mathbf{R}}_j$ . Since  $L$  is quadratic and diagonal in the velocities,

$$\mathbf{p}_i = \frac{\partial L}{\partial \dot{\mathbf{r}}_i} = m_e \dot{\mathbf{r}}_i \Rightarrow \dot{\mathbf{r}}_i = \frac{\mathbf{p}_i}{m_e}, \quad \mathbf{P}_j = \frac{\partial L}{\partial \dot{\mathbf{R}}_j} = M_j \dot{\mathbf{R}}_j \Rightarrow \dot{\mathbf{R}}_j = \frac{\mathbf{P}_j}{M_j}.$$

**1.3 Legendre transform (D).**  $H = \sum_i \mathbf{p}_i \cdot \dot{\mathbf{r}}_i + \sum_j \mathbf{P}_j \cdot \dot{\mathbf{R}}_j - L$ . Substituting  $\dot{\mathbf{r}}_i = \mathbf{p}_i / m_e$ ,  $\dot{\mathbf{R}}_j = \mathbf{P}_j / M_j$ :

approximated,

approximated ones

## ■ Some Other Tricks

- Before diving into derivations, ask Claude what I should learn first
- Add a dependency tag — e.g., Eq. (1.5) comes from Eq. (1.2) and Eq. (1.3)
- Make Claude use **SymPy** to partially automate verification steps
- Add built-in sanity checks on the dimensional analysis of key formulas
- Independent re-derivation: Let Claude derive the same result via two different methods

# ■ Conclusion

- Your own judgement is critical
- **Context** is extremely important
- Formulate your problem as close to a math problem as possible  
(Give all definitions, symbols, initial & final points)
- Pack repetitive steps & instructions as Agent Skills
- **Thanks to the AI Library for AI subscription support!**

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<https://forms.gle/JuowfH6v29CgWaqv7>



# Thank you

Questions & discussion welcome

# Recent Progress of AI in Computer Science

- Donald Knuth's article

## Claude's Cycles

Don Knuth, Stanford Computer Science Department  
(28 February 2026; revised 14 April 2026)

Shock! Shock! I learned yesterday that an open problem I'd been working on for several weeks had just been solved by Claude Opus 4.6—Anthropic's hybrid reasoning model that had been released three weeks earlier! It seems that I'll have to revise my opinions about "generative AI" one of these days. What a joy it is to learn not only that my conjecture has a nice solution but also to celebrate this dramatic advance in automatic deduction and creative problem solving. I'll try to tell the story briefly in this note.

1974 recipient of the  
ACM Turing Award



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[The Yoda of Silicon Valley](#)  
- [The New York Times](#)

# Recent Progress of AI in Math Research

- Timothy Gowers' weblog

## A recent experience with ChatGPT 5.5 Pro

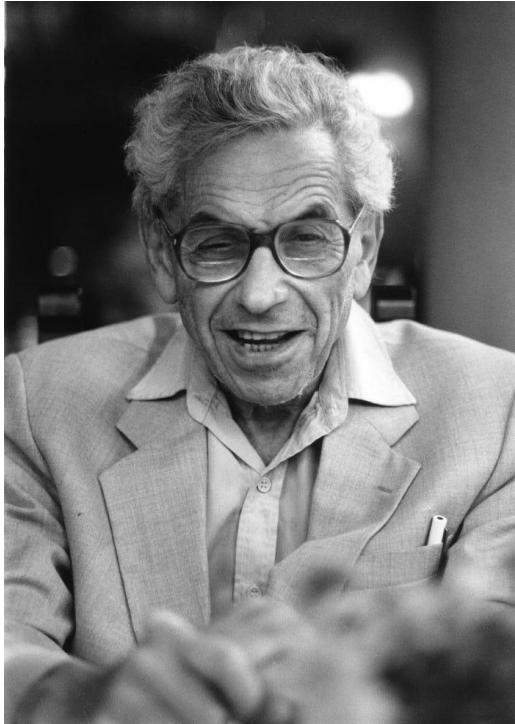
We are all having to keep revising upwards our assessments of the mathematical capabilities of large language models. I have just made a fairly large revision as a result of ChatGPT 5.5 Pro, to which I am fortunate to have been given access, producing a piece of PhD-level research in an hour or so, with no serious mathematical input from me.

1998 Fields Medalist



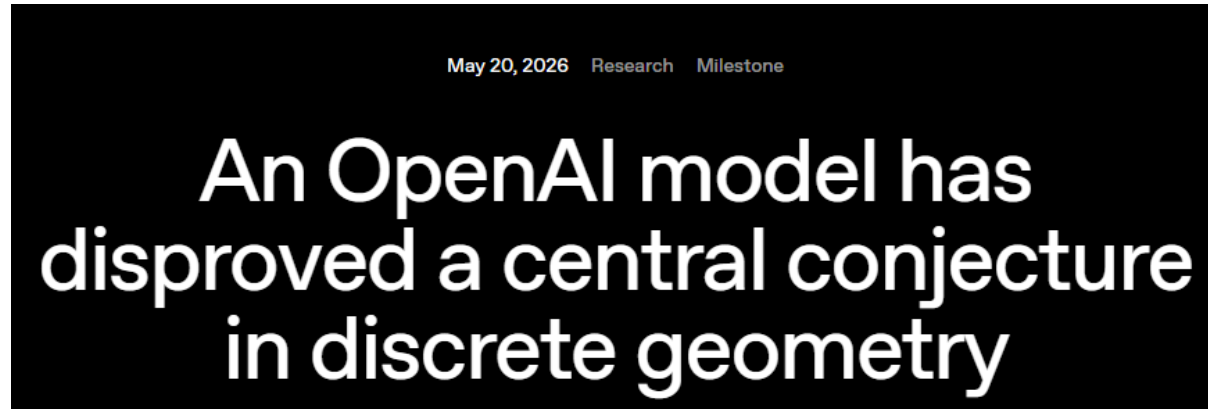
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# Recent Progress of AI in Math Research



Paul Erdős

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Of Highest Intellectual Calibre](#)



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[An OpenAI model has disproved a central conjecture in discrete geometry](#)

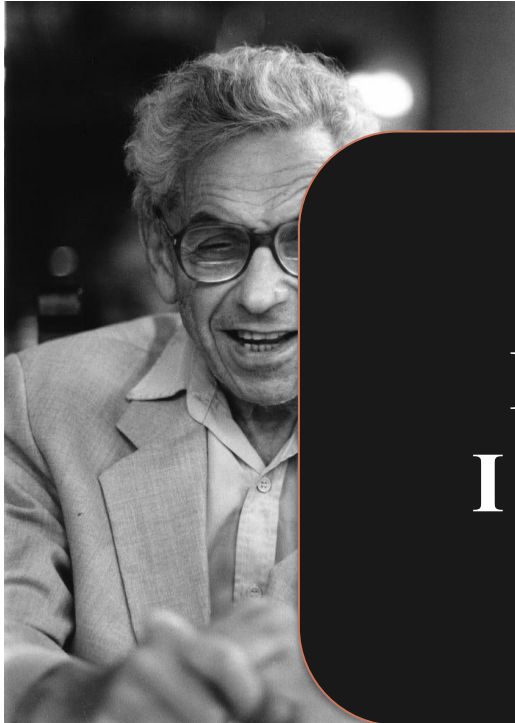
## REMARKS ON THE DISPROOF OF THE UNIT DISTANCE CONJECTURE

NOGA ALON, THOMAS F. BLOOM, W. T. GOWERS, DANIEL LITT, WILL SAWIN, ARUL SHANKAR, JACOB TSIMERMAN, VICTOR WANG, AND MELANIE MATCHETT WOOD

ABSTRACT. We present a short, digested, human-verified version of the recent OpenAI-generated counterexample to the Erdős unit distance conjecture, and a sequence of reflections on it. The argument relies crucially on ideas that may, at least in retrospect, be attributed to Ellenberg-Venkatesh, Golod-Shafarevich, and Hajir-Maire-Ramakrishna.

[arXiv:2605.20695](#)

# Recent Progress of AI in Math Research



**Inspired by these breakthroughs,  
I should use AI in my own research**

Paul Erdős

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May 20, 2026 Research Milestone

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[OpenAI model has  
proved a central conjecture  
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ANCE CONJECTURE

SAWIN, ARUL SHANKAR,  
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